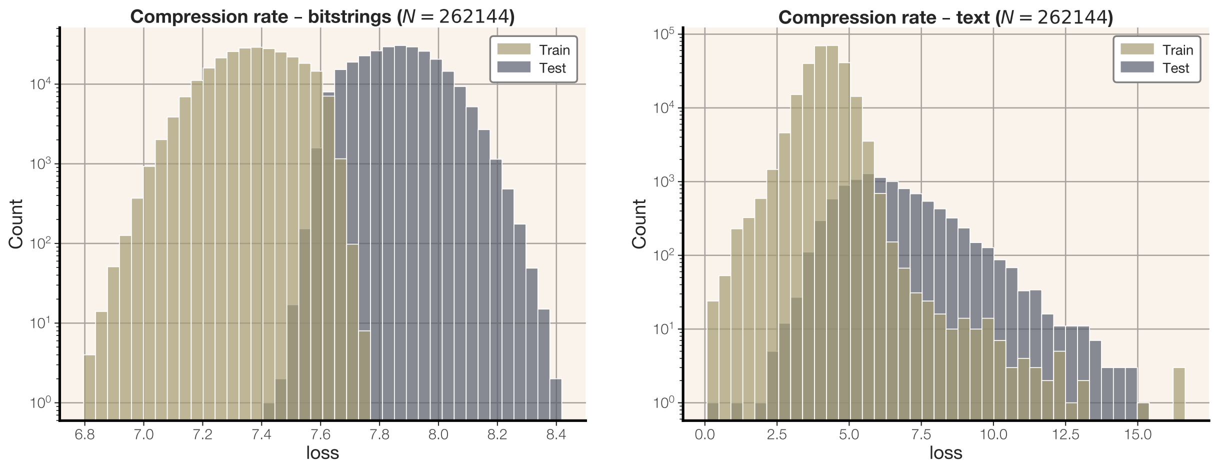

Figure 15 Distribution of compression rates for equal-sized transformers ($n_{\text{layer}} = 4$, $d_{\text{model}} = 128$) trained on 2^{14} sequences of equal-length random bitstrings (left) and text (right).

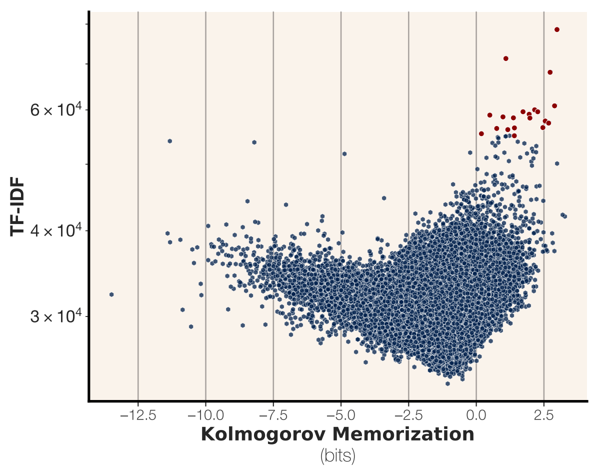

Figure 16 Unintended memorization vs. TF-IDF for all training points of a $20M$ param model trained past its capacity on 2^{16} sequences of English text. The training documents with rarest words are typically the most memorized.

$$\text{TF-IDF}(d; \mathcal{D}) = \frac{1}{|d|} \sum_{w \in d} \log \frac{|D|}{tf(w, \mathcal{D})}$$

where $tf(d, \mathcal{D})$ indicates the total number of times word w appears in dataset $\mathcal{D}$. Intuitively, a higher TF-IDF score for document d indicates that d contains more words that are rare in $\mathcal{D}$.

We clearly observe for samples with positive unintended memorization there is a strong correlation between trainset TF-IDF and memorization: examples with more rare words are more memorized. In particular, the sample with highest TF-IDF out of the whole training dataset (a sequence of Japanese words) has the third-highest measured memorization; even though this is just one out of 260,000 training samples, the model can regurgitate the entire sequence given just a single token (囚). Out of the top twenty memorized sequences, all but three contain sequences of tokens from other languages (Japanese, Chinese, and Hebrew).

Manual analysis (Table 5) indicates that the most memorized datapoints have extremely rare tokens, typically ones not found in English.

	Text	TFIDF	Memorization	Language
0	人気エリアであるフォンニャに位置するRock & Roll Hostelは、ビジネス出張と観光のどちらにも最適なロケーションです。�	78553.72	2.98	Japanese
1	このトピックには0件の返信が含まれ、1人の参加者がいます。1 年、 6 ヶ月前に Dave Gant さんが最後の更新	71279.19	1.09	Japanese
2	Label: Living Records\nDestroy All MonstersメンバーBen Millerによる自主レーベルからのソロCD-R。こちらは付属の抽象画をサウンド化したと	68064.46	2.73	Japanese
3	《左傳》記「崔氏側莊公于北郭。丁亥，葬諸士孫之里，四翣，不�	60820.46	2.89	Chinese
4	歡迎客人自備紋身圖案或要求本紋身店代客起圖， 設計起圖須	60018.53	2.16	Chinese
5	By 小森 栄治,向山 洋一\nRead Online or Download 中学の理科「総まとめ」を7日間で攻略する本 「�	59625.40	2.27	Japanese
6	統合分析是將一些議題相關但彼此獨立的臨床實驗之研究結果(大�	59624.37	1.73	Chinese
7	在SIA-Smaart Pro的Real-Time Module实时模块上，将功能扩展，实时显示相位和Fixed Point Per Octave（	59128.54	1.95	Chinese
8	Progress in Intelligent Transportation Systems and IoT/M2M Communications: Markets, Standardization, Technologies\n|出版日||ページ情報||英文 173 Pages\n|インテリジェント交通システムお	58953.67	0.50	Japanese
9	English Title: Kingdom Hearts: Chain of Memories\nJapanese Title: キングダム ハーツ チェイン オブ メモリーズ — “Kingdom Hearts: Chain of Memories”\nAuthor: Tomoco Kanemaki\nIllustrator: Shiro	58605.92	0.99	Japanese
10	אשכול זה הועבר לארכיון. נא לשאול שאלה חדשה אם יש לך צורך	58420.30	1.37	Hebrew
11	在《易經》里单数为阳, 双数为阴. 我曾怀疑马来西亚政府也会	58382.40	1.98	Chinese
12	「XXI c.—21世紀人」第3回企画展 三宅一生ディレクション\n21_21 DESIGN SIGHT 第 3 回企画展の	57797.99	2.55	Japanese
13	无敌神马在线观看 重装机甲 睿峰影院 影院 LA幸福剧本\n时间：2020-12	57399.24	2.67	Chinese
14	季末小邪 回复 dgutkai: 楼主 您好 可以把项目源码发我吗？ 可以付�	56539.93	2.46	Chinese
15	�בכל סדנא אפשר לזהות את הילדים שהוריהם מאפשרים חופש יצי	56478.18	1.41	Hebrew
16	Ακαδημαϊκές Δημοσιεύσεις Μελών ΔΕΠ σε άλλα Ιδρύματα >\n�	56376.74	0.75	Greek
17	Larry想和李华，还有她那些中国朋友多在一起玩儿，了解更多的中国文	56152.72	1.16	Chinese
18	Mark 5:18 wrote:καὶ ἐμβαίνοντος αὐτοῦ εἰς τὸ πλοῖον παρεκάλει αὐ	55391.28	0.19	Greek
19	בתחילה הייתי סקפטית לגבי השקעת כסף בשיווק אינטרנטי.	55014.00	1.41	Hebrew

Table 5 Highest TF-IDF training examples from a $20M$ param model trained past its capacity on 2^{16} sequences of English text. All of the highest TF-IDF examples are considered memorized, and contain text from non-English languages (Japanese, Chinese, Hebrew, and Greek).

A.4 Scaling law fit

Here we demonstrate the fit of our sigmoidal scaling law to experimental data. We show points in tokens-per-parameter vs. fit in Figure 17. Although the sigmoidal function is slightly simplistic (the points do not perfectly fit the curve) our fit produces estimates within $1 - 2\%$ of observations.

A.5 Proofs

In the section we provide the proofs missing from the main body.

A.6 Proof of Proposition 1

Here we prove Proposition 1

Proof. we have

$$\begin{aligned}\mathrm{mem}_U(X, \hat{\Theta}, \Theta) &= I(X \mid \Theta, \hat{\Theta}) \\ &= I((X_1 \mid \Theta, \ldots, X_n \mid \Theta), \hat{\Theta}).\end{aligned}$$

And since the data is sampled i.i.d., all random variables in $\{R_i = [X_i \mid \Theta]\}_{i \in [n]}$ are independent. [4] So we have,

$$I((X_1 \mid \Theta, \ldots, X_n \mid \Theta), \hat{\Theta}) \geq \sum_{i \in [n]} I(X_i \mid \Theta, \hat{\Theta})$$

[4]Note that X_i themselves are not independent because they are sampled by first sampling an underlying model Θ. However, they are conditionally independent once the underlying model Θ is given.

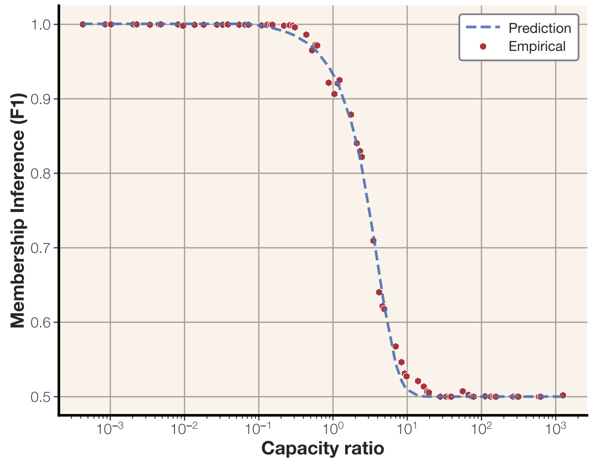

Figure 17 Our sigmoidal scaling law for membership inference fit to experimental data.

which implies

$$\mathrm{mem}_U(X, \hat{\Theta}, \Theta) \geq \sum_{i \in [n]} \mathrm{mem}_U(X_i, \hat{\Theta}, \Theta).$$

On the other hand, we have

$$\begin{aligned}\mathrm{mem}_U(X, \hat{\Theta}, \Theta) &= I(X \mid \Theta, \hat{\Theta}) \\ &= H(\hat{\Theta} - H(\hat{\Theta} \mid (X \mid \Theta)) \\ &\leq H(\hat{\Theta})\end{aligned}$$

□

A.7 Proof of Proposition 4

Proof. We first state a Lemma about connection between algorithmic (kolmogorov) mutual information and mutual information.

Lemma 6. [Theorem 3.6 in Grunwald & Vitányi (2004)] Assume (X, Y) be a pair of joint random variables. Let f be the density function, $f(x, y) = \Pr[(X, Y) = (x, y)]$. Then we have

$$\begin{aligned}I(X, Y) - H_K(f) &\leq \mathop{\mathrm{E}}_{(x,y)\sim(X,Y)}[I_K(x, y)] \\ &\leq I(X, Y) + 2H_K(f).\end{aligned}$$

Now we use this lemma to prove the statement of the Proposition. Let f be a the density function for the joint distribution $(X_i \mid \theta, \hat{\Theta})$. That is $f_i(x_i, \hat{\theta}) = \Pr[X_i = x_i \mid \theta \text{ and } \hat{\Theta} = \hat{\theta}]$. Note that this function is independent of n. By definition we have

$$\mathrm{mem}_U(X_i, \hat{\Theta}, \theta) = I(X_i \mid \theta, \hat{\Theta}).$$

Now using Lemma 6 we have

$$\begin{aligned}I(X_i \mid \theta, \hat{\Theta}) - H_K(f) &\leq \mathop{\mathrm{E}}_{x_i \sim X_i \mid \theta}[I_K(x_i, \hat{\theta})] \\ &\leq I(X_i \mid \theta, \hat{\Theta}) + 2H_K(f).\end{aligned}$$

and this conclude the statement of Proposition by setting $\epsilon = 2H_K(f)$ □

A.8 Limitations

Our efforts to measure language model memorization come from a line of recent research to discover whether models have analyzed certain texts, and if so, how much. However, our main experimental contributions relate to the practice of training and evaluating language models, including a new perspective on the phenomenon of grokking (Nakkiran et al., 2019) and a new measurement of capacity. Our results are specific to the environment proposed and do not necessarily generalize to other datasets, architectures, or training setups.

B Discussion of other notions of memorization

In this section we list multiple other notions of memorization and compare it with our definition. We specifically focus on why these notions do not satisfy all of our requirements.

- **Stability-based notions of memorization.** There are notions of privacy and memorization that deal with "stability" of the training algorithm to small changes in the training set. Most notably, differential privacy Dwork (2006) considers the worst-cast drift of the model distribution when a single data point changes. Another notion of memorization in Feldman (2020) is based on the change of the model prediction on a point x, when we add the labeled pair (x, y) to the training set of a classification/regression model. Both of these notions are crucially relying on the learning algorithm and how it behaves. Moreover, the definition of differential privacy is not ideal for our case because it is a worst-case definition and cannot be applied at sample/model level. While the notion of memorization in Feldman (2020) does not have this particular issue, it suffers from the fact that it only applies to classification models and mostly deals with the memorization of the association between the label (y) and input (x), and not the memorization of x itself. These issues make these notions not ideal for our case.

- **Extraction-based memorization.** There are multiple works in the literature (Carlini et al., 2019; Mireshghallah et al., 2022; Nasr et al., 2023; Zhang et al., 2023; Carlini et al., 2023b; Schwarzschild et al., 2024) that define memorization of samples in language models based on how easy it is to extract that sample. Specifically, when trying to understand the extent of memorization of a sample x in a model θ they measure some notion of complexity for the task of eliciting the model to output x. Although these notions are great in that they only take a model θ and a sample x, they still do not account for generalization. Considering our running example of the following training sample: "What is 2^{100}? (A: $1, 267, 650, 600, 228, 229, 401, 496, 703, 205, 376$)", this will be identified as highly memorized by almost all of the extraction based notions of memorization. Another issue with these definitions are that they are heavily dependent on the details of decoding algorithm. This is not ideal as we do not expect the memorization of a sample x in a model θ to depend on the detailed parameters we use to generate samples using θ.

  The work of Schwarzschild et al. (2024) in this category is the closest to ours. This work which is based on prompt-optimization, optimizes a short prompt p to make the model elicit x, then it calls the sample x memorized, if length of p is less than x. Although this definition is close to our definition in using compression, it still does not account for generalization of the model. Moreover, it focuses on a specific way of compression through prompting. We posit that compression through prompting is an inferior compression scheme and can often lead to compression rates greater than 1.

- **Membership/attribute inference.** Membership inference Shokri et al. (2017) and attribute inference attacks Jayaraman & Evans (2022) have been used for empirically measuring the privacy of machine learning algorithms. These notions which usually aim at approximating the stability notions of memorization are suffering from the same shortcomings. They rely heavily on the learning algorithm and the data distribution. Moreover, they fail at providing a sample level notion of memorization. For example, the obtained accuracy for membership inference attack is only meaningful in the population level. This is because various attack may have different true positives for membership, and the union of all these true positive across different attack may cover the entire training set, rendering it unusable as a sample level notion of memorization.

- **Data copying in generative models.** There are some interesting notions of memorization designed specifically for generative modeling where a generative model may output a certain portion of training samples